

Simulation study: mixture of Wishart

The Tien Mai & Zhi Zhao

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Data simulation:

- Sample size $n = 200$
- Dimension of the Wishart distribution $p = 10$
- Number of latent components $K = 5$
- Probabilities of subpopulations $\boldsymbol{\pi} = (0.35, 0.10, 0.25, 0.2, 0.1)$
- Degrees of freedom $\boldsymbol{\nu} = (p + 5, p + 2, p + 10, p + 1, p + 3)$
- Scale matrices of the Wishart distribution $\Sigma_k, k = 1, \dots, K$
- Data $S_i \sim \sum_{k=1}^K \pi_k \text{Wishart}(\nu_k, \Sigma_k)$

Modeling:

- Bayesian mixture of five Wishart distributions

```
rm(list = ls())

library(moewishart)

set.seed(123)
n <- 200 # number of subjects
p <- 10

dat <- simData(n, p, K = 5,
               pis = c(0.35, 0.10, 0.25, 0.2, 0.1),
               nus = c(p+5, p+2, p+10, p+1, p+3),
               Sigma = NULL # default Sigmas pre-defined in function 'simData()'
)
S_list <- dat$S
Sigma_list <- dat$Sigma_list

my_K_intial <- 5

# run a moewishart model
start_time <- Sys.time()
set.seed(123)
fit <- moewishart(S_list,
                  K = my_K_intial, mh_sigma = 1,
                  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE
)
```

```
Iter    1 | LL=-31637.1 | 57.0 iter/sec | acc_rate_nu=[0.000 1.000]
Iter  500 | LL=-26792.7 | 467.8 iter/sec | acc_rate_nu=[0.010 0.030]
Iter 1000 | LL=-26817.6 | 467.1 iter/sec | acc_rate_nu=[0.012 0.030]
Iter 1500 | LL=-26802.0 | 465.8 iter/sec | acc_rate_nu=[0.012 0.029]
Iter 2000 | LL=-26819.5 | 466.8 iter/sec | acc_rate_nu=[0.011 0.033]
```

```

Iter 2500 | LL=-26802.7 | 469.3 iter/sec | acc_rate_nu=[0.010 0.030]
Iter 3000 | LL=-26802.7 | 471.5 iter/sec | acc_rate_nu=[0.009 0.030]
Iter 3500 | LL=-26802.7 | 472.5 iter/sec | acc_rate_nu=[0.010 0.031]
Iter 4000 | LL=-26801.2 | 471.8 iter/sec | acc_rate_nu=[0.009 0.030]
Iter 4500 | LL=-26825.5 | 471.4 iter/sec | acc_rate_nu=[0.010 0.031]
Iter 5000 | LL=-26815.0 | 471.4 iter/sec | acc_rate_nu=[0.010 0.029]
Iter 5500 | LL=-26803.6 | 473.0 iter/sec | acc_rate_nu=[0.010 0.028]
Iter 6000 | LL=-26811.4 | 472.3 iter/sec | acc_rate_nu=[0.010 0.028]
Iter 6500 | LL=-26840.4 | 472.2 iter/sec | acc_rate_nu=[0.010 0.028]
Iter 7000 | LL=-26805.0 | 473.2 iter/sec | acc_rate_nu=[0.011 0.029]
Iter 7500 | LL=-26811.2 | 474.1 iter/sec | acc_rate_nu=[0.011 0.029]
Iter 8000 | LL=-26818.3 | 474.4 iter/sec | acc_rate_nu=[0.011 0.030]
Iter 8500 | LL=-26800.8 | 475.0 iter/sec | acc_rate_nu=[0.011 0.030]
Iter 9000 | LL=-26829.8 | 475.2 iter/sec | acc_rate_nu=[0.012 0.030]
Iter 9500 | LL=-26803.7 | 475.2 iter/sec | acc_rate_nu=[0.012 0.030]
Iter 10000 | LL=-26819.2 | 475.5 iter/sec | acc_rate_nu=[0.012 0.030]

```

```

end_time <- Sys.time()
elapsed_time <- end_time - start_time
elapsed_time

```

Time difference of 21.05127 secs

```

# posterior summaries
colMeans(fit$pi)

```

```
[1] 0.34596297 0.13146322 0.06354975 0.23932287 0.21970119
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.35 0.10 0.25 0.20 0.10
```

```
apply(fit$nu, 2, mean)
```

```
[1] 15.05522 12.50074 10.85340 19.42432 11.06481
```

```
dat$nu # true nu
```

```
[1] 15 12 20 11 13
```

EM algorithm

```

# run the model with EM algorithm
set.seed(1234)
fit3 <- moewishart(S_list, method = "em",
                  K = my_K_intial, mh_sigma = 1,
                  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE
)

```

```

Iter 10 | Loglik: -27619.9279 | Nu: 15, 13.43, 11.11, 11.71, 19.91
Iter 20 | Loglik: -27619.8208 | Nu: 15, 13.44, 11.11, 11.71, 20.11
Converged at iteration 27

```

```

# EM estimates
fit3$pi

```

```
[1] 0.35 0.13 0.22 0.06 0.24
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.35 0.10 0.25 0.20 0.10
```

```
fit3$nu
```

```
[1] 14.99616 13.43616 11.11132 11.70752 20.11382
```

```
dat$nu # true nu
```

```
[1] 15 12 20 11 13
```